FFSD PROJECT

(1st Evaluation Documentation)

Domain : Smart City & Urban Services Platforms

Problem Statement : Urban Service Request & Issue Management Platform

Team Name : Urbanity

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Software Requirements Specification (SRS)

Urbainity – Urban Service Request & Issue Management Platform

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1. Preface

This document is intended for project evaluators, faculty members, developers, and other stakeholders involved in the design, development, and assessment of the Urban Service Request and Issue Management Platform. It provides a structured and comprehensive description of the system requirements to ensure a shared understanding among all readers.

This Software Requirements Specification (SRS) document corresponds to version 1.0 of the project. The primary objective of this version is to capture the complete set of functional and non-functional requirements of the system at an early stage of development. It serves as a baseline for system design, implementation, testing, and future enhancements.

This version has been created to formally document system expectations, user interactions, and operational constraints identified during the initial requirements analysis phase. Future versions of this document may incorporate refinements based on stakeholder feedback, changes in project scope, or evolving system needs.

2. Introduction

2.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define the functional and non-functional requirements of the Urban Service Request and Issue Management Platform. This document serves as a common reference for developers, project mentors, evaluators, and stakeholders to understand the system's objectives, expected behavior, and constraints.

2.2 Scope of the Project

The Urban Service Request and Issue Management Platform is a web-based application developed to address the need for efficient handling of civic issues in urban environments. The system enables citizens to report public service issues, track their status, and receive updates, while allowing municipal authorities to manage, assign, and resolve complaints in a structured manner.

The platform supports Smart City objectives by promoting transparent governance, improved service delivery, and active citizen participation through digital means. It provides centralized monitoring, accountability, and data-driven insights for urban service management.

The system may interact with external services such as mapping systems for location-based complaint reporting, notification services for status updates, and dashboard components for analytics and performance monitoring. It is designed to integrate smoothly with existing urban administration workflows while remaining scalable for future enhancements.

3. Glossary

Term	Description
Citizen	A registered user who submits complaints or service requests
Complaint	A reported civic issue related to public infrastructure or services
Service Request	A formal request raised by a citizen for an urban service
Issue Category	Classification of a complaint based on the type of problem
Priority Level	Indicates the urgency of a complaint (Low, Medium, High)
Status	The current state of a complaint (Submitted, Assigned, In Progress, Resolved, Closed)
Department Officer	Official responsible for reviewing and managing complaints
Field Worker	Personnel assigned to inspect or resolve complaints on the ground
Department Head	Authority overseeing departmental operations and performance
Admin	System administrator managing users, roles, and system settings
Assignment	The process of allocating a complaint to an officer or field worker
Escalation	Forwarding an unresolved complaint to a higher authority
Resolution	Action taken to successfully address and close a complaint
Feedback	Response provided by a citizen after issue resolution

Notification	System-generated alert sent to users regarding updates
Dashboard	Interface displaying complaint statistics and system status
Location Data	Geographic information associated with a complaint
Resolution Time	Duration from complaint submission to closure

4. Description

4.1 Product Perspective

The system is a standalone web application developed using a full-stack architecture. It consists of a frontend interface for users, a backend server for business logic and processing, and a database for persistent data storage. The platform may integrate with external services such as map APIs, notification services, or analytics tools.

4.2 User Classes and Characteristics

User Type	Description
Citizen	Submits complaints and tracks their status
Department Officer	Reviews and assigns complaints
Field Worker	Performs on-ground resolution tasks
Department Head	Monitors department performance
Admin	Manages system settings and user roles

4.3 Operating Environment

The platform is accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Brave, Safari, and Microsoft Edge. No special software installation is required. The application runs on a centralized server or cloud infrastructure, with all data securely stored in a database system.

4.4 Design and Implementation Constraints

- Internet connectivity is required to access the system
- The platform must comply with basic data security practices
- System performance depends on server and network availability

5. User Requirements

5.1 Citizen Requirements

- Citizens shall be able to register and log in securely
- Citizens shall be able to submit complaints with descriptions and location details
- Citizens shall be able to view the status of their submitted complaints
- Citizens shall receive notifications on status updates
- Citizens shall be able to provide feedback after issue resolution

5.2 Department Officer Requirements

- Officers shall be able to log in securely
- Officers shall view and manage complaints assigned to their department
- Officers shall assign complaints to field workers
- Officers shall update the complaint status

5.3 Field Worker Requirements

- Field workers shall view the assigned tasks
- Field workers shall update progress and resolution details
- Field workers shall upload proof of work if required

5.4 Department Head Requirements

- Department heads shall monitor complaint statistics
- Department heads shall review departmental performance
- Department heads shall escalate unresolved issues

5.5 Admin Requirements

- Admin shall manage users and roles
- Admin shall configure system settings
- Admin shall monitor overall system activity

6. System Architecture

The Urban Service Request and Issue Management Platform follows a modular, web-based architecture designed to support scalability, maintainability, and clear separation of responsibilities. The system is organized into distinct components, each responsible for a specific set of functions.

The **frontend module** provides the user interface for all system users, including citizens, department officers, field workers, department heads, and administrators. It enables users to interact with the system for activities such as complaint submission, status tracking, task assignment, and monitoring through dashboards. The frontend is accessible through standard web browsers and focuses on usability and responsiveness.

The **backend services** handle the core application logic and workflow management. These services process user requests, manage authentication and authorization, control the complaint lifecycle, and enforce business rules such as assignment and escalation. Backend services act as the central coordination layer between the user interface, the database, and external services.

The **database component** is responsible for persistent storage of system data, including user profiles, complaints, assignments, status updates, and feedback. It ensures data consistency, integrity, and reliable retrieval of information required by the application.

The system supports **external integrations** to enhance functionality and user experience. Mapping services may be used to capture and visualize location-based complaint data. Notification services may be used to deliver updates to users regarding complaint status changes. Dashboard and analytics components may leverage reusable visualization libraries or services.

Where appropriate, existing and proven services such as authentication mechanisms and map APIs may be reused to reduce development effort and improve reliability. The architecture is designed to allow future integration with additional services without major structural changes.

7. System Requirements

- Role-based access control(RBAC) must be supported for all users.
- User inputs must be validated before processing or storage.
- Duplicate complaints for the same issue and location should be prevented.
- Important system actions must be recorded in audit logs.
- Authorized users should be able to update the complaint priority and category.
- Complaint escalation should be supported based on predefined rules.
- Notifications must be generated for important status updates.
- Timestamps must be maintained for complaint creation, assignment, and resolution.
- Administrative users should be able to enable or disable platform features.
- Search and filter options must be available for managing complaints.
- Complaint submissions should support attachments such as images or documents.
- Data consistency must be maintained across all modules.
- Recovery mechanisms should be available to handle system failures.
- Basic analytics should be provided to monitor complaint trends.
- The architecture should allow future integration with external services.

8. Functional Requirements

8.1 User Registration Function

- The system provides a registration function that allows new users to create an account by submitting required identification details.

8.2 User Authentication Function

- The system provides an authentication function that verifies user credentials and establishes authorized access to the platform.

8.3 Role-Based Access Function

- The system provides role-based access functionality that enables or restricts features based on assigned user roles.

8.4 Complaint Submission Function

- The system provides a complaint submission function that allows citizens to register civic issues with relevant details.

8.5 Complaint Assignment Function

- The system provides an assignment function that routes submitted complaints to the appropriate department.

8.6 Complaint Status Management Function

- The system provides a status management function that updates and tracks the progress of complaints throughout their lifecycle.

8.7 Monitoring Dashboard Function

- The system provides monitoring dashboard functionality for authorized officials to view system and complaint metrics.

8.8 Reporting Function

- The system provides reporting functionality to generate summaries related to complaint resolution and system activity.

9. Non-Functional Requirements

9.1 Performance

- The system shall respond within acceptable time limits
- The system shall support multiple concurrent users

9.2 Security

- User passwords shall be encrypted
- Secure communication using HTTPS
- Unauthorized access shall be prevented

9.3 Usability

- The system shall be easy to use for non-technical users
- The interface shall be responsive for desktop and mobile devices

9.4 Reliability

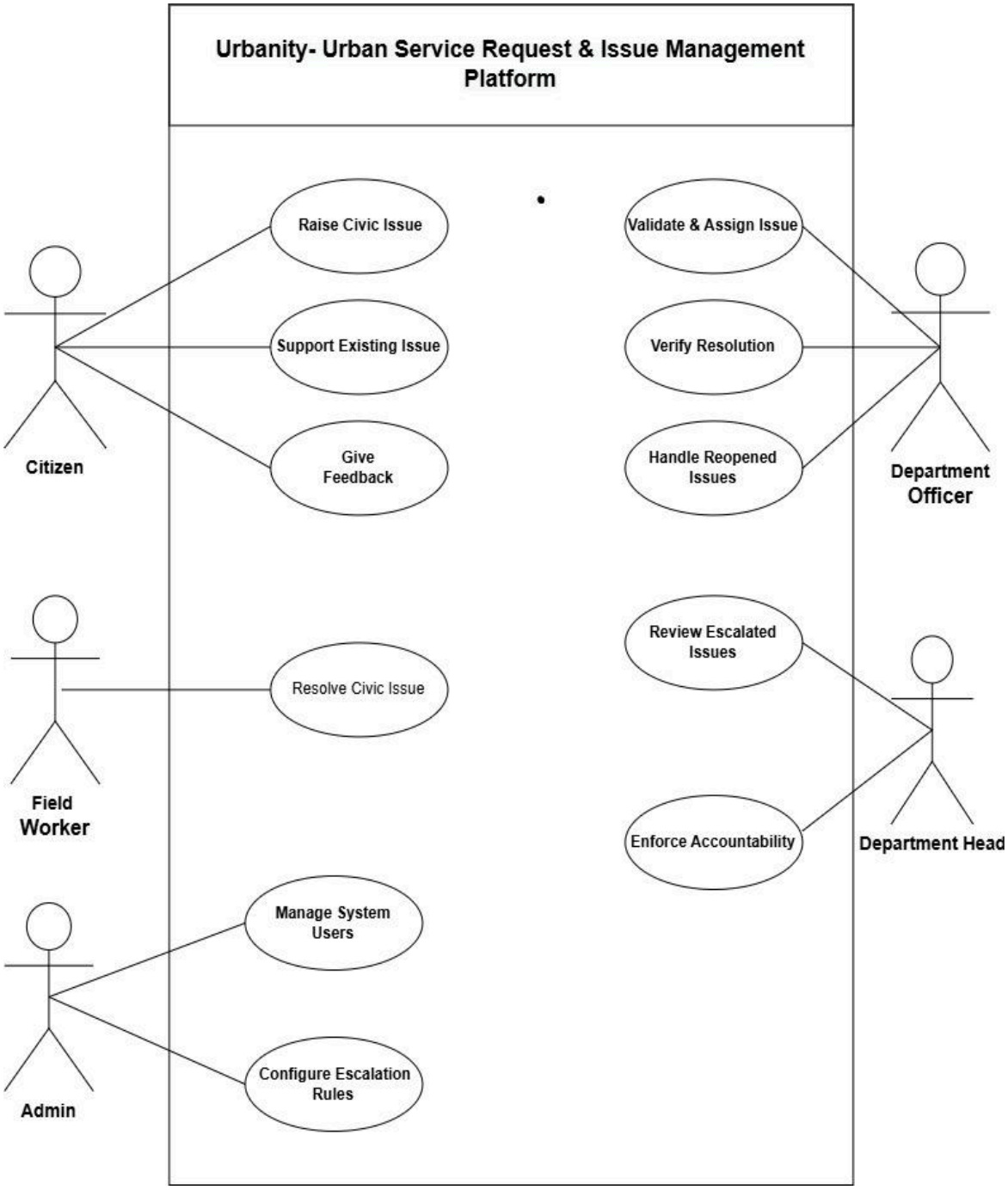
- The system shall ensure high availability
- The system shall handle failures gracefully

9.5 Scalability

- The system shall support future feature enhancements
- The system shall handle increased user load

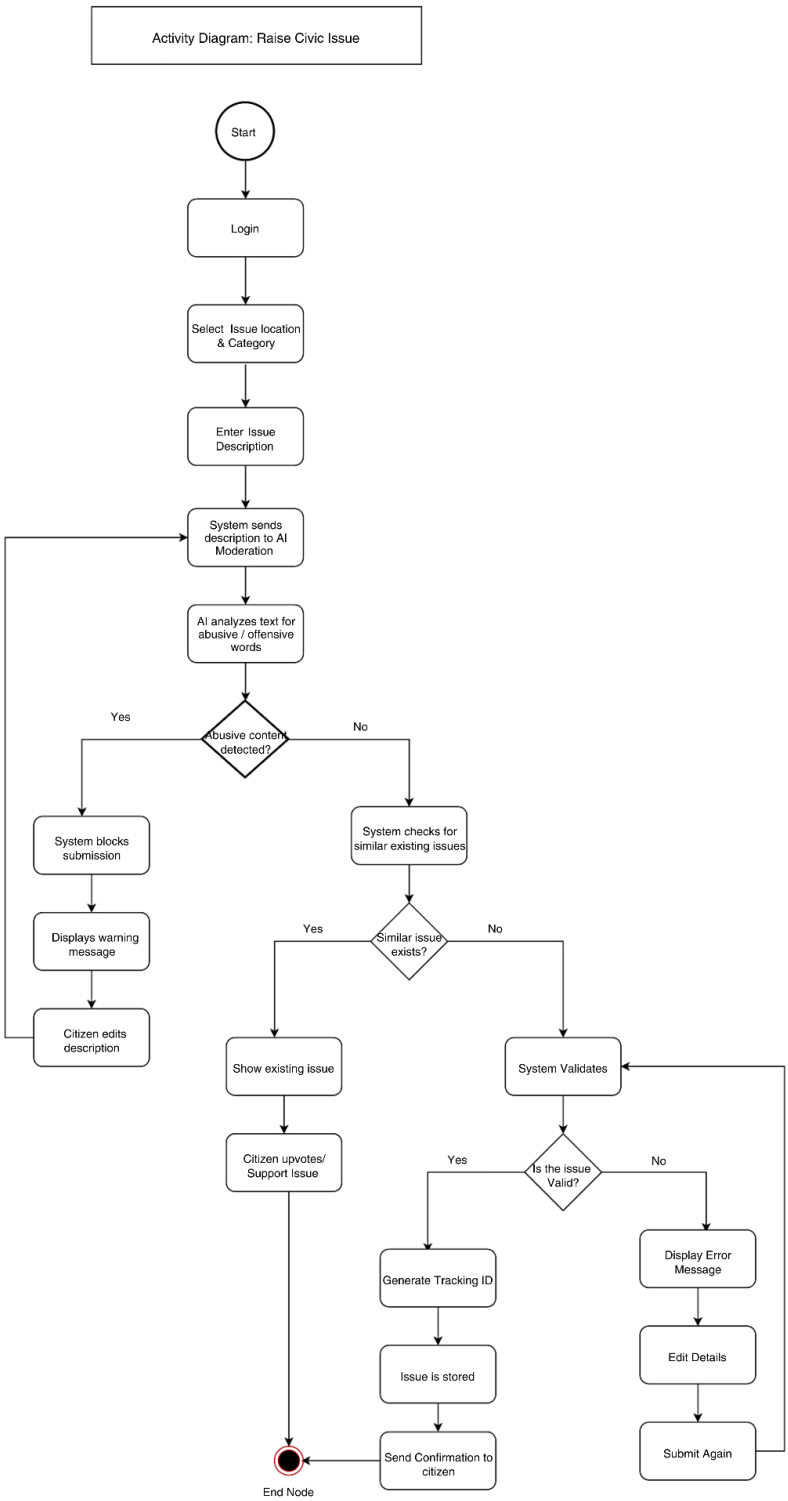
10. System Models

10.1 Use Case Diagram:

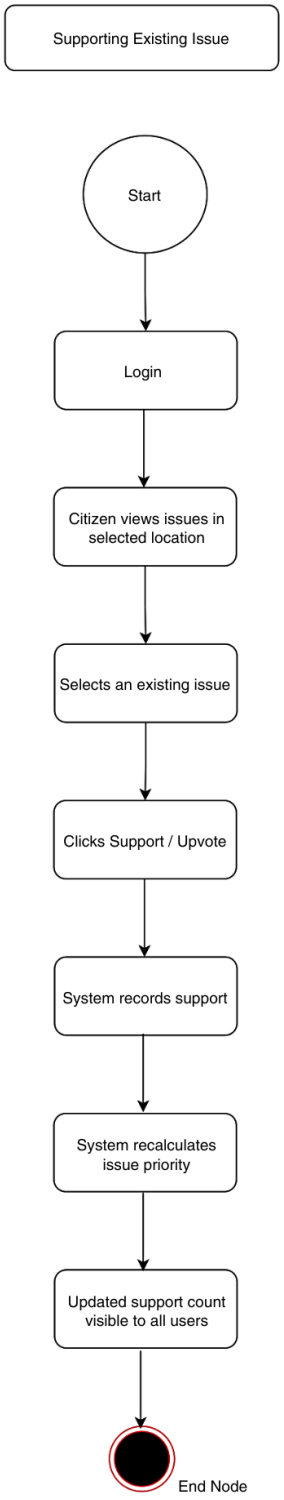


10.2 Activity Diagrams for major workflows:

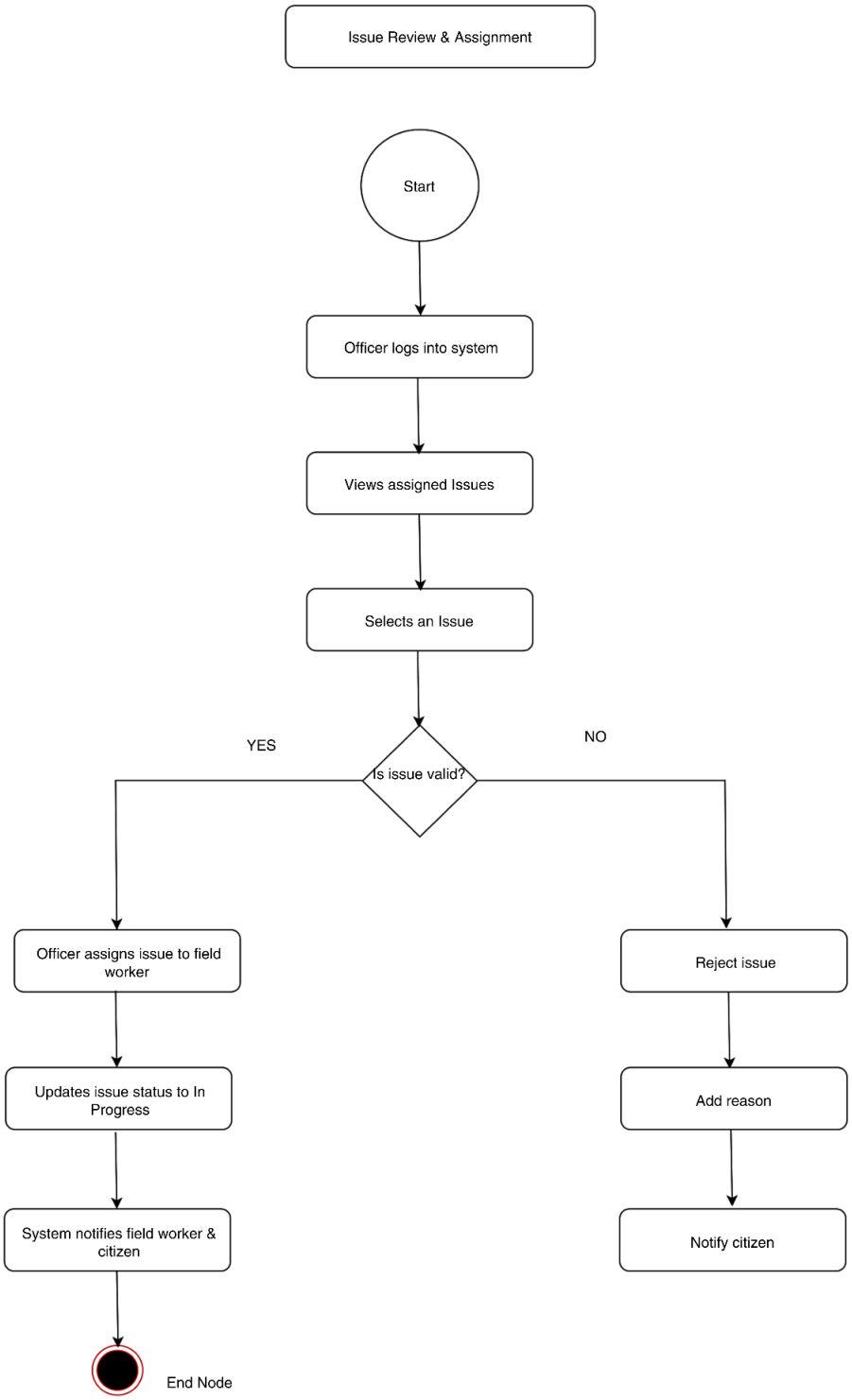
1) Raise an Issue(Citizen):



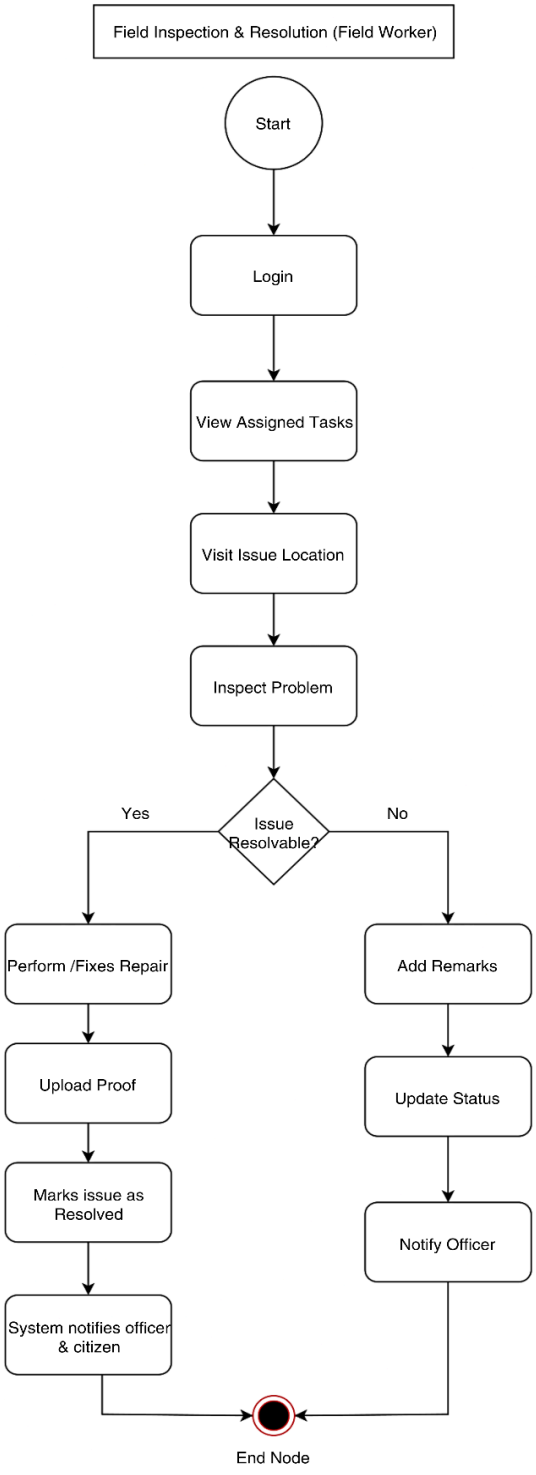
2) Support / Upvote Existing Issue (Citizen):



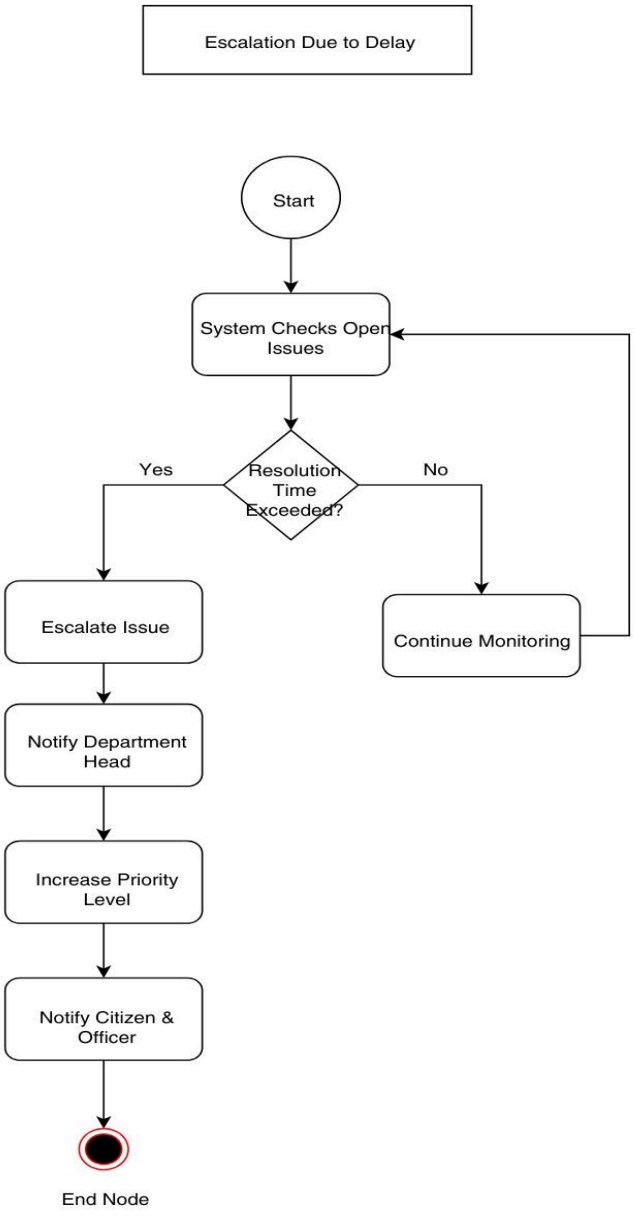
3) Issue Assignment & Review (Department Officer):



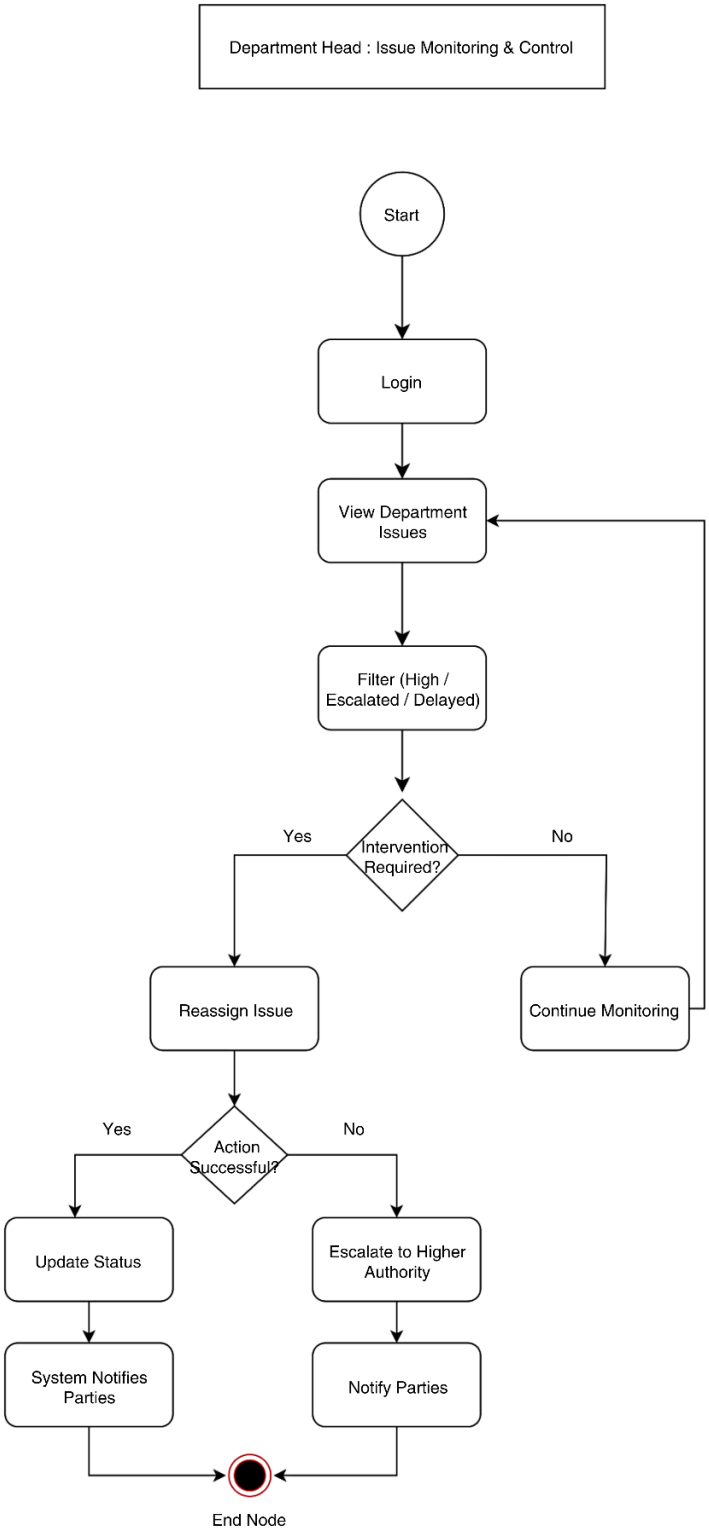
4) Field Inspection & Resolution(Field Worker):



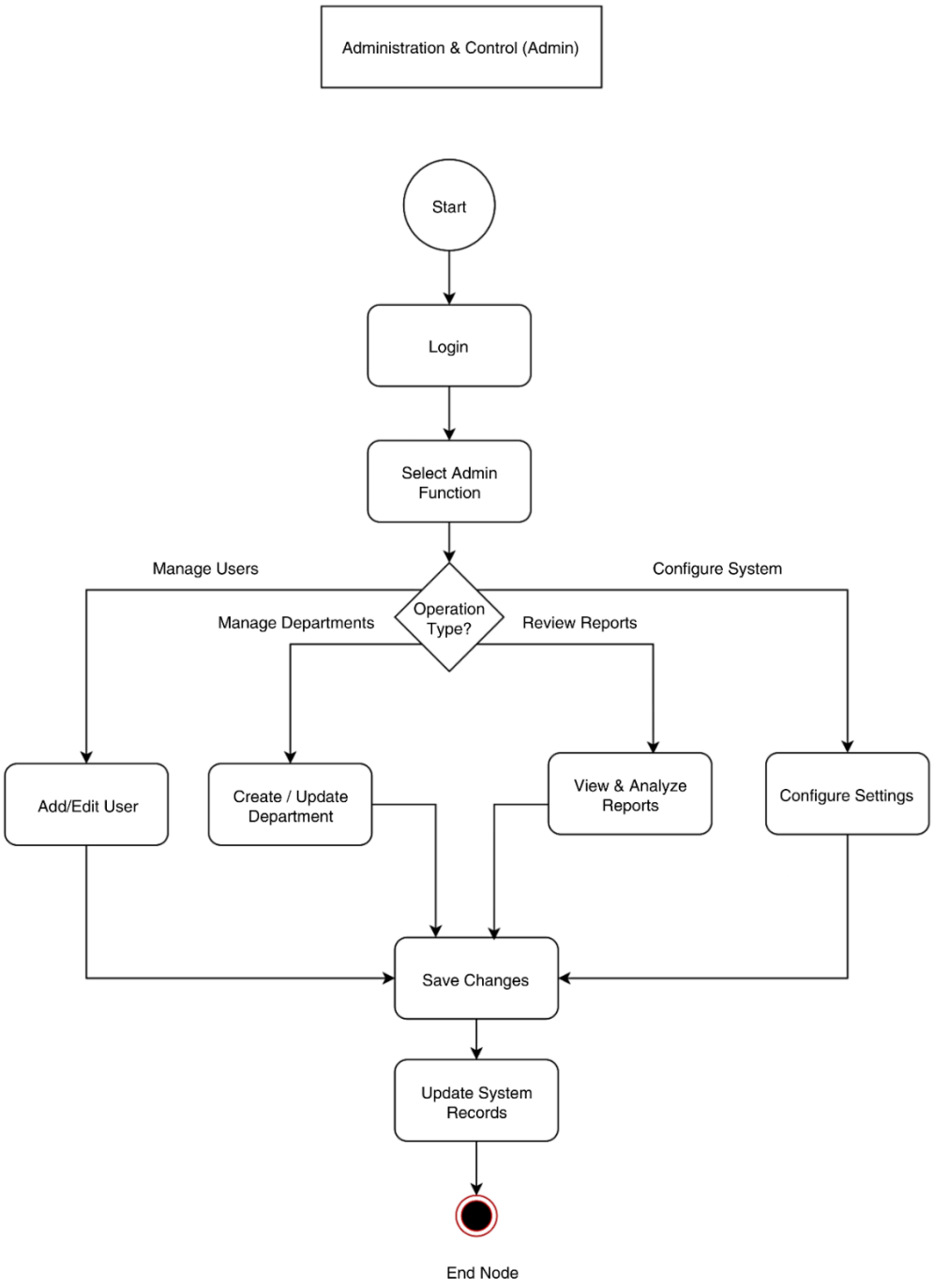
5) Escalation Due to Delay (System + Dept Head):



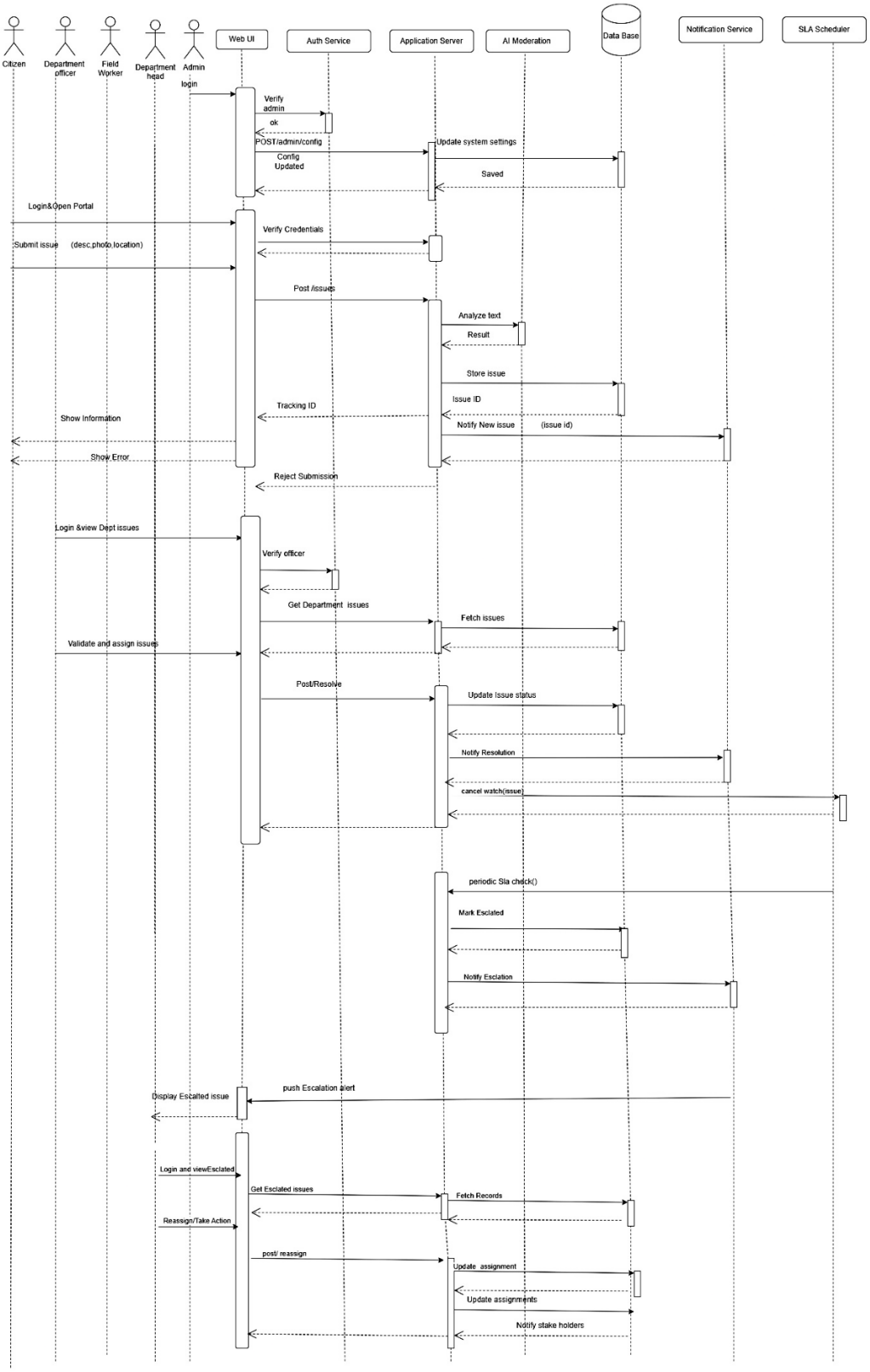
6) Issue Monitoring & Control (Department Head):



7) User & System Management (Admin):



10.3 Sequence Diagram:



These diagrams visually represent user interactions, system processes, and data flow.

11. System Evolution

The Urban Service Request and Issue Management Platform is designed based on the assumption that it will operate as a web-based system accessible through standard browsers. It assumes stable internet connectivity for users and server-side infrastructure capable of supporting concurrent access.

As urban service requirements evolve, the system is expected to undergo enhancements to support additional departments, services, and user roles. Future changes may include the introduction of mobile applications, integration with advanced analytics systems, and support for automated decision-making based on historical data.

The system architecture is designed to accommodate changes in user volume, service scope, and integration with new external services without requiring major restructuring. Anticipated changes in technology, such as improvements in mapping services, notification mechanisms, or security standards, can be incorporated with minimal impact on existing functionality.

12. Appendices

This section provides supplementary information related to the Urban Service Request and Issue Management Platform that supports, but does not form part of, the core system requirements.

Appendix A: Hardware Requirements

The system requires a device with internet connectivity and a modern web browser for end users. On the server side, a standard web server environment with sufficient processing power, memory, and storage is required to support concurrent users and data storage.

Appendix B: Database Overview

The system maintains structured data related to users, complaints, assignments, status updates, and feedback. Data relationships are organized to ensure consistency, integrity, and efficient retrieval. Detailed database schemas may be defined during the design and implementation phase.

Additional appendices may be added in future versions of this document to include detailed technical specifications or deployment-related information.

13. Index

This document includes multiple indexes to assist readers in locating information efficiently. A general alphabetical index is provided to reference key terms, concepts, and sections used throughout the document.

In addition, separate indexes are maintained for diagrams and functional elements. The diagram index lists all visual models, such as use case diagrams, activity diagrams, sequence diagrams, and feature diagrams, included in the document. The function index references major system functions and workflows described in the requirements sections.

These indexes improve navigability and readability, especially for evaluators and developers who need to quickly access specific information. Additional indexes may be introduced in future versions of the document if required.

--Thank You--